National University of Computer and Emerging Sciences
Lahore Campus

Database Systems Lab(CL2005)

Date: May20th 2026

Instructor(s)

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Final Exam

Total Time (Hrs):	**2**
Total Marks:	**80**
Total Questions:	**2**

Roll No | Section | Student Signature

Do not write below this line

Submission Path

Windows + R: \\Exam\Lab Submissions\DB Lab Final\ [your section]

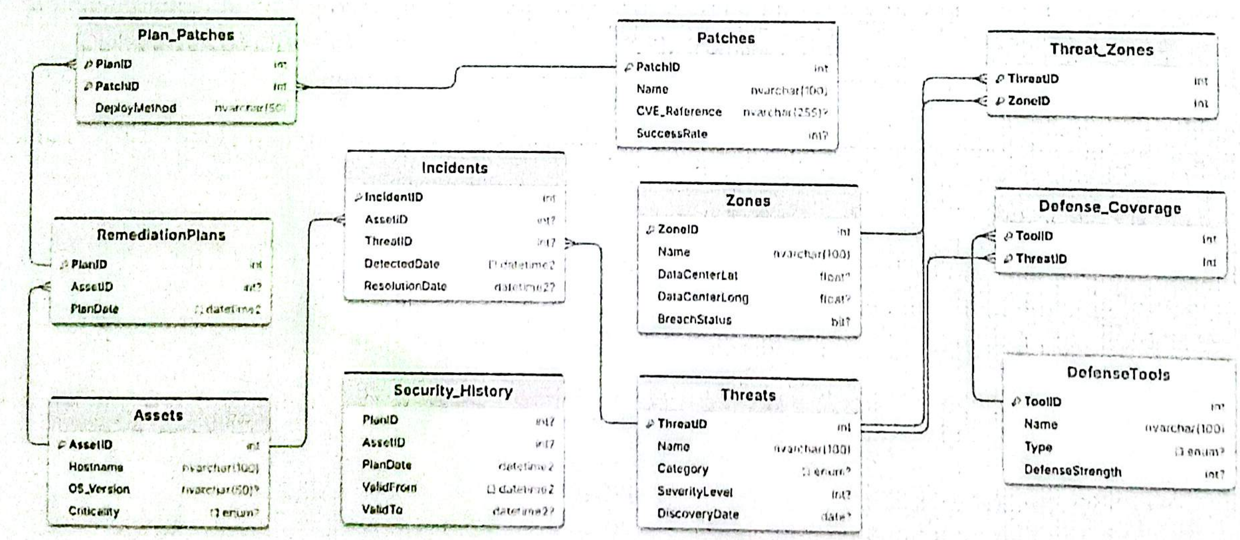
Important Note

- Submit a **single .sql file** named using your roll number e.g.**l23-1234.sql.**
- Clearly mention the Question and the part you are answering in **a comment**
- After submitting, **verify** your file is visible in the **submission folder of your section**, and **check that a file size is shown** next to it (to ensure it's not an empty 0 KB file)

Attempt all the questions.

CyberShield DB

CyberShield is a cybersecurity management framework that mirrors biological disease tracking to defend digital infrastructure. It identifies and maps digital threats across network segments, tracks active security breaches on critical hardware, and coordinates the deployment of software patches and defensive tools to neutralize risks.



0259

Q1 [40 Marks]

a) **[5 Marks]** Write a DDL statement to ensure that `CVE_Reference` in Patches table has a format 'CVE-{4 digits}-{3 digits}'. eg: 'CVE-2025-101'

b) **[5 Marks]** Write a DDL statement to ensure that `DeployMethod` in Plan_Patches contains 'deployed' after a hyphen. Eg: 'static-method-deployed'

c) **[10 Marks]** Create a view **vw_ThreatSummary**. Display the threat category as `threat_category`, their average severity level as `avg_severity`, total count of unresolved incidents (`ResolutionDate` is null) as `total_incidents`, number of distinct assets for each threat category as `total_assets`. Only show threat categories which have at least 3 unresolved incidents in the current month (`DetectedDate` is in current month) OR have affected an asset with `criticality` either `High` or `Mission-Crit`.

d) **[10 Marks]** Write an **AFTER UPDATE** trigger **tr_ZonalBreachStatus** on the Threats table. The trigger must set `BreachStatus` = 1 for all associated zones in the Zones table if the updated threat's `SeverityLevel` becomes strictly greater than the maximum `DefenseStrength` of the defense tools fighting it. Otherwise, turn off the `BreachStatus` bit.

e) **[10 Marks]** Write a stored procedure named **sp_FinalizeIncidents** that accepts a @PlanID from a `RemediationPlan` and runs a **Transaction.** In which, it finds every unresolved incident of the associated asset from the given remediation plan, then updates the incident's `ResolutionDate` to the current date, only for incidents whose `DiscoveryDate` is earlier than the RemediationPlan's `PlanDate`. Finally, it increments the `SuccessRate` in Patches by 1 for all patches deployed under the remediation plan. Use appropriate savepoints and Try Catch blocks. Rollback and print relevant error messages in the catch block if the transaction fails.

Q2 [40 Marks]

a) **[5 Marks]** Write a DDL statement to ensure that `Name` in Zones table doesn't contain 'Zone' in any case. Eg: Both '101-zoNe-A' and '103_ZOnE' are invalid.

b) **[5 Marks** Write a DDL statement to ensure that `Name` in DefenseTools has 'tool' in any case, at the beginning or at the end. Eg: 'anti-worm-tool' or 'Tool-anti-worm'

c) **[10 Marks]** Create a view named **vw_AssetVulnerabilityProfile** that displays every asset's `Hostname`, total number of logged incidents as `number_of_incidents`, and the cumulative hours exposed to security threats as `hours_exposed`. The cumulative exposure must sum the hour differences between `DetectedDate` and `ResolutionDate` for resolved entries, and use the current date as the terminal endpoint for active, unresolved incidents, with the total aggregate hours rounded down to the nearest integer. All assets must be shown and order by `hours_exposed` in descending order.

d) **[10 Marks]** Write an INSTEAD OF UPDATE trigger named **tr_RestrictWeekendResolution** on the Incidents table. The trigger must block any updates to the `ResolutionDate` column if the current date falls on a weekend (Saturday or Sunday) **AND** all deployment methods (DeployMethod) currently linked to that asset's remediation plans contain the phrase 'manual compile', otherwise allow the update to proceed.

e) **[10 Marks]** Write a stored procedure named **sp_DeployFirewallDefenses** that accepts a @ZoneID and executes a transaction using a TRY...CATCH engine. First it finds every threat in the zone and verifies that it is **attacked by every** `Category` of Threat. Then only for those zones attacked by all categories of threats, it tries to attach a `Firewall` type DefenseTool to every threat affecting assets with a criticality of either `High` or `Mission-Crit`. A `Firewall` type DefenseTool must n't be associated with more than 5 threats, so using more than one `Firewall` type tool is allowed. In case there are **no remaining** `Firewall` tools, **rollback** the transaction. Use appropriate savepoints. Rollback and print relevant error messages in the catch block if the transaction fails.